

Total No. of Questions : 8]

SEAT No. :

P711

[Total No. of Pages : 2

[6004]-709

B.E. (E&TC) (Honors)

ARTIFICIAL INTELLIGENCE IN ROBOTICS
(2019 Pattern) (Semester - VIII) (404183HR)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Use of Calculator is allowed.
- 5) Assume Suitable data if necessary.

- Q1)** a) Write Heuristics search techniques in AI. [5]
b) Explain various types of uniformed search algorithm. [5]
c) What is real coded genetic algorithm. [8]

OR

- Q2)** a) What is simulated annealing? [5]
b) Write in details about informed search algorithm. [5]
c) Write in details about ant colony optimization. [8]

- Q3)** a) What is image based robot guidance? [5]
b) Write about application of image processing in robotics. [5]
c) Write in detail about machine vision algorithm. [8]

OR

- Q4)** a) Explain image based camera exploration for autonomous robot. [5]
b) Write in details about types of image processing. [5]
c) Explain automatic sorting and inspection algorithm. [8]

- Q5)** a) What is mobile robot motion planning? [5]
b) What is autonomous robot hand? [5]
c) Explain with block diagram path planning robot. [7]

OR

- Q6)** a) What is task based hybrid closure grasping? [5]
b) Explain accurate motion control for fast mobile robot. [5]
c) Explain with block diagram obstacle avoidance robot. [7]

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- Q7)** a) What is tool management in AI in flexible automation? [5]
b) Explain applications of various intelligent system in AI in flexible automation. [5]
c) What is process planning in flexible automation? [7]

OR

- Q8)** a) Write short note on Real time scheduling. [5]
b) Explain tool management in detail. [5]
c) Write short note on route optimization for AS/RS systems. [7]

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